NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 9: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. In the context of a Markov Decision Process (MDP), which sequence best describes the basic idea of Value Iteration?

A. Compute state utilities → select optimal actions
B. Select a policy → compute its utilities → stop
C. Select actions randomly → update rewards
D. Compute rewards → construct a fixed policy
Correct Answer: (A)

Q2. In a Partially Observable Markov Decision Process (POMDP), the agent should base its decisions primarily on:

A. The most recent percept alone
B. The complete sequence of past actions
C. A probability distribution over possible states
D. A fixed sequence of predetermined actions

Correct Answer: (C)

Q3. The value of a policy is defined as the expected cumulative sum of _____ obtained while following the policy.

A. Immediate rewards
B. Discounted rewards
C. Maximum rewards
D. Terminal rewards
Correct Answer: (B)

Q4. Consider the state sequence $[S_0, S_1, S_2]$ with rewards

$$R(S_0) = 10,\ R(S_1) = 4,\ R(S_2) = 2$$

and discount factor

$$\gamma = 0.5.$$

Using the discounted-reward formulation, what is the utility of the state sequence?

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A. 12.00
B. 12.50
C. 14.00
D. 16.00

Correct Answer: (B)
Detailed Solution:
The utility of the sequence using the discounted-reward formulation,

$$U([S_0, S_1, S_2]) = R(S_0) + \gamma R(S_1) + \gamma^2 R(S_2)$$
$$U = 10 + (0.5 * 4) + (0.5^2 * 2) = \mathbf{12.50}$$

Q5. An agent considers two possible states, S_1 and S_2, with equal prior probabilities:

$$P(S_1) = P(S_2) = 0.5.$$

An agent maintains a belief distribution over possible states.
A sensor produces percept E, with:

$$P(E \mid S_1) = 0.8,\ P(E \mid S_2) = 0.2.$$

After receiving percept E, what is the updated belief

$$P(S_1 \mid E)?$$

A. 0.80
B. 0.60
C. 0.50
D. 0.20

Correct Answer: (A)
Detailed Solution:
Using Bayes' rule to find the posterior belief P(S1|E):

$$P(S_1|E) = \frac{P(E|S_1)P(S_1)}{P(E)}$$
$$P(E) = P(E|S_1)P(S_1) + P(E|S_2)P(S_2)$$
$$P(E) = (0.8 * 0.5) + (0.2 * 0.5) = 0.50$$
$$P(S_1|E) = \frac{0.40}{0.50} = \mathbf{0.80}$$

Q6. An agent in state S can choose either A_1 or A_2. The one-step action values are:

$$Q(S, A_1) = 5.2$$
$$Q(S, A_2) = 4.7.$$

Which action should the agent select under the current utility estimates?

A. Select A_1 because its action value is higher
B. Select A_2 because its action value is lower
C. Select either action because both are feasible
D. Select neither action because values are estimated

Correct Answer: (A)

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Q7. In Policy Iteration, suppose that after Policy Evaluation, the current policy π_0 is found to select A_1 in state S, but policy improvement shows that A_2 has a higher expected utility in S.
What should the algorithm do next?

A. Retain A_1 and terminate immediately
B. Replace A_1 with A_2 in the policy
C. Randomly select between A_1 and A_2
D. Recalculate all transition probabilities

Correct Answer: (B)

For Q8 – Q10

Consider the following MDP: An agent is currently in state S. It can choose either A_1 or A_2.
For the purposes of one-step Value Iteration, the successor states are A and B.

Quantity	**State** A	**State** B
Immediate reward $R(s')$	2	0
Current utility $U(s')$	4	2

The discount factor is. $\gamma = 0.5$.
The transition probabilities are:

Action	$P(A' \mid Sa)$	$P(B' \mid Sa)$
A_1	0.75	0.25
A_2	0.25	0.75

The one-step discounted action value is

$$Q(S,a) = \sum P(s' \mid S,a)[R(s') + \gamma U(s')].$$

Q8. For action A_1, what is the **expected immediate reward** over the successor states?

A. 1.50
B. 1.75
C. 2.00
D. 2.50

Correct Answer: (A)
Detailed Solution:
The expected immediate reward is

$$E[R|S,A_1] = P(A|S,A_1)R(A) + P(B|S,A_1)R(B)$$
$$E[R|S,A_1] = (0.75 * 2) + (0.25 * 0) = \mathbf{1.50}$$

Q9. Using the given reward, utility and transition values, what is the **one-step discounted action value** for A_1?

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A. 2.75
B. 3.00
C. 3.25
D. 3.50

Correct Answer: (C)
Detailed Solution:
Successor state A: R(A) = 2, U(A) = 4

$$R(A) + \gamma U(A) = 2 + 0.5(4) = 4$$

Successor state B: R(B) = 0, U(B) = 2

$$R(B) + \gamma U(B) = 0 + 0.5(2) = 1$$

Transition probabilities for A_1:

$$P(A|S, A_1) = 0.75, P(B|S, A_1) = 0.25$$

$$Q(S, A_1) = P(A|S, A_1)[R(A) + \gamma U(A)] + P(B|S, A_1)[R(B) + \gamma U(B)]$$

$$Q(S, A_1) = 0.75 * [2 + 0.5(4)] + 0.25 * [0 + 0.5(2)] = \mathbf{3.25}$$

Q10. Using the same MDP, **which action should be selected** under the current utility estimates?

A. A_1, because $Q(S, A_1) > Q(S, A_2)$
B. A_2, because $Q(S, A_2) > Q(S, A_1)$
C. Either action, because both have equal utility
D. Neither action, because the MDP is stochastic
Correct Answer: (A)

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